SQL:

This work focuses on analyzing organizational data to generate meaningful insights about employees, departments, and company-wide trends. Using SQL queries across core tables such as EMPLOYEES, DEPARTMENTS, JOBROLES, SALARIES, PERFORMANCE_REVIEWS, and LOCATIONS1, the analysis highlights key aspects of workforce management and decision-making.

Database

USE COMPANY_DB3;

Tables

DEPARTMENT
Departments define the organizational structure by grouping employees into functional units. Each department is identified by a unique ID and name, allowing HR teams to evaluate performance, attrition, and stability at the unit level. This table is crucial for comparing workforce metrics across different business areas and identifying which departments are thriving or struggling.

CREATE TABLE DEPARTMENTS(DEPID INT PRIMARY KEY,DNAME VARCHAR(35) NOT NULL UNIQUE);

LOCATIONS1
The locations table records the geographical distribution of employees, linking each individual to a specific office or region. This information is valuable for location-wise salary comparisons, diversity analysis, and understanding workforce distribution. It helps organizations assess whether compensation and representation are balanced across different regions.

CREATE TABLE LOCATIONS1(LID INT PRIMARY KEY,LNAME VARCHAR(35) NOT NULL);

JOBROLES
The job roles table outlines the positions available within the company. Each role is linked to employees through a role ID, making it possible to track career progression and promotions. By analyzing job roles, HR can understand salary structures, identify employees who have advanced through multiple roles, and evaluate readiness for leadership positions.

CREATE TABLE JOBROLES(RID INT PRIMARY KEY,RNAME VARCHAR(35) NOT NULL,MIN_SALARY DECIMAL(10,2),MAX_SALARY DECIMAL(10,2));

EMPLOYEES
The employees table serves as the central hub of the database, containing essential information about each individual in the organization. It typically includes identifiers such as employee ID and name, along with

references to their department, job role, and location. Additional attributes like join date, employment status, and gender (if tracked) provide context for tenure, diversity, and workforce stability.

```
CREATE TABLE EMPLOYEES(EMPID INT PRIMARY KEY,ENAME VARCHAR(35) NOT NULL,GENDER VARCHAR(11),DEPID INT,RID INT,LID INT,JOINDATE DATE,STATUS VARCHAR(35) DEFAULT 'ACTIVE',FOREIGN KEY(DEPID) REFERENCES DEPARTMENTS(DEPID),FOREIGN KEY(RID) REFERENCES JOBROLES(RID),FOREIGN KEY(LID) REFERENCES LOCATIONS1(LID));
```

SALARIES

Salary data is stored in this table, linked to employees by their ID. It provides the basis for compensation analysis, enabling comparisons across departments, roles, and locations. Salary information also supports advanced analytics such as correlation studies between pay and performance, and helps identify cases where compensation may not align with contribution.

```
CREATE TABLE SALARIES(SALID INT PRIMARY KEY,EMPID INT,SALARY DECIMAL(10,2),EFFECTIVE_DATE DATE,FOREIGN KEY(EMPID) REFERENCES EMPLOYEES(EMPID));
```

PERFORMANCE_REVIEWS

Performance reviews capture employee ratings over time, usually by year. This table is essential for monitoring trends, evaluating promotion eligibility, and identifying high performers who may be suitable for leadership roles. It also enables department-level performance comparisons and supports correlation analysis with salary data.

```
CREATE TABLE PERFORMANCE_REVIEWS(REVIEWID INT PRIMARY KEY,EMPID INT,REVIEW_YEAR INT,RATING DECIMAL(2,1) CHECK (RATING BETWEEN 1 AND 5),FOREIGN KEY(EMPID) REFERENCES EMPLOYEES(EMPID));
```

INSERTING VALUES TO TABLES DIV

```
BULK INSERT DEPARTMENTS
FROM 'C:\Dataset\Departments Dataset - Sheet1.csv'
WITH (
   FIRSTROW = 2,
   FIELDTERMINATOR = ',',
   ROWTERMINATOR = '\n',
   TABLOCK
);

BULK INSERT LOCATIONS1
FROM 'C:\Dataset\Locatioin_dataset - Sheet1.csv'
WITH (
   FIRSTROW = 2,
   FIELDTERMINATOR = ',',
```

```
    ROWTERMINATOR = '\n',
    TABLOCK
);

BULK INSERT JOBROLES
FROM 'C:\Dataset\Job_roles_dataset - Sheet1.csv'
WITH (
    FIRSTROW = 2,
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '\n',
    TABLOCK
);

BULK INSERT EMPLOYEES
FROM 'C:\Dataset\Employees_100_Records - Employees_100_Records.csv'
WITH (
    FIRSTROW = 2,
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '\n',
    TABLOCK
);

BULK INSERT SALARIES
FROM 'C:\Dataset\Salaries_Data - Salaries_Data.csv'
WITH (
    FIRSTROW = 2,
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '\n',
    TABLOCK
);

BULK INSERT PERFORMANCE_REVIEWS
FROM 'C:\Dataset\performance.xlsx - Sheet1.csv'
WITH (
    FIRSTROW = 2,
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '\n',
    TABLOCK
);
```

PERFORMANCE ANALYSIS

1.TOP10 PERFORMANCE OF EMPLOYEES .

SELECT TOP 10 EMPLOYEES.* FROM PERFORMANCE_REVIEWS inner join EMPLOYEES ON PERFORMANCE_REVIEWS.EMPID=EMPLOYEES.EMPID ORDER BY PERFORMANCE_REVIEWS.RATING DESC;

Results | Messages

	EMPID	ENAME	GENDER	DEPID	RID	LID	JOINDATE	STATUS
1	108	Rohan Gupta	Male	3	2	1	2022-11-28	Active
2	153	Amit Kumar	Male	3	2	2	2019-01-02	Active
3	115	Arjun Iyer	Male	2	4	3	2023-06-24	Inactive
4	171	Sneha Kumar	Female	1	1	1	2017-10-06	Active
5	196	Rohan Patel	Male	5	5	1	2017-01-27	Inactive
6	192	Divya Rao	Female	2	3	1	2019-04-29	Active
7	169	Rahul Gupta	Male	3	1	1	2017-12-21	Active
8	154	Meera Kumar	Female	2	5	2	2017-08-15	Inactive
9	134	Naveen Rao	Male	5	4	3	2023-08-23	Active
10	142	Pooja Rao	Female	1	1	2	2023-02-23	Inactive

2.Average rating per department

SELECT D.DNAME AS Department, AVG(P.RATING) AS AvgRating FROM EMPLOYEES E JOIN DEPARTMENTS D ON E.DEPID = D.DEPID JOIN PERFORMANCE_REVIEWS P ON E.EMPID = P.EMPID GROUP BY D.DNAME ORDER BY AvgRating DESC;

Results | Messages

	Department	AvgRating
1	HR	3.094444
2	DotNET	2.935000
3	Data	2.915789
4	DevOps	2.878260
5	Java	2.730000

3.Employees eligible foR promotion (rating>4 & 2+ years)

SELECT E.EMPID, E.ENAME, D.DNAME AS Department, J.RNAME AS JobRole, P.RATING, E.JOINDATE, DATEDIFF(YEAR, E.JOINDATE, GETDATE()) AS YearsInCompany
FROM EMPLOYEES E JOIN DEPARTMENTS D ON E.DEPID = D.DEPID
JOIN JOBROLES J ON E.RID = J.RID
JOIN PERFORMANCE_REVIEWS P ON E.EMPID = P.EMPID WHERE P.RATING > 4 AND DATEDIFF(YEAR, E.JOINDATE, GETDATE()) >= 2 ORDER BY P.RATING DESC, YearsInCompany DESC;

Results | Messages

	EMPID	ENAME	Department	JobRole	RATING	JOINDATE	YearsInCompany
2	108	Rohan Gupta	DevOps	SDE 1	5.0	2022-11-28	4
3	171	Sneha Kumar	HR	HR	4.9	2017-10-06	9
4	196	Rohan Patel	DotNET	Manager	4.9	2017-01-27	9
5	115	Arjun Iyer	Data	Data Scientist	4.9	2023-06-24	3
6	154	Meera Kumar	Data	Manager	4.8	2017-08-15	9
7	169	Rahul Gupta	DevOps	HR	4.8	2017-12-21	9
8	192	Divya Rao	Data	Associate	4.8	2019-04-29	7
9	134	Naveen Rao	DotNET	Data Scientist	4.8	2023-08-23	3
10	142	Pooja Rao	HR	HR	4.7	2023-02-23	3
11	117	Kavya Shar...	DevOps	Associate	4.6	2016-04-05	10
12	180	Neha Rao	DotNET	SDE 1	4.5	2015-05-18	11
13	173	Swathi Rao	DevOps	SDE 1	4.5	2017-12-25	9
14	116	Swathi Rao	DotNET	Data Scientist	4.5	2017-05-11	9
15	157	Meera Verma	Java	SDE 1	4.3	2016-12-28	10

4.PERFORMANCE TREND OVER YEARS

SELECT REVIEW_YEAR, AVG(RATING) AS AvgPerformanceRating, COUNT(*) AS TotalReviews FROM PERFORMANCE_REVIEWS GROUP BY REVIEW_YEAR ORDER BY REVIEW_YEAR;

	REVIEW_YEAR	AvgPerformanceRating	TotalReviews
1	2015	2.844444	9
2	2016	3.116666	6
3	2017	3.566666	3
4	2018	3.325000	8
5	2019	3.388888	9
6	2020	3.100000	6
7	2021	2.200000	9
8	2022	3.030000	10
9	2023	2.921428	14
10	2024	2.328571	7
11	2025	2.814285	7

SALARY ANALYTICS

5.SALARY DISTRIBUTION PER DEPARTMENT

SELECT E.DEPID,SUM(S.SALARY) FROM SALARIES S INNER JOIN EMPLOYEES E ON S.EMPID=E.EMPID GROUP BY E.DEPID;

	DEPID	(No column name)
1	1	12440000.00
2	2	13840000.00
3	3	17300000.00
4	4	15010000.00
5	5	16160000.00

6.HIGHEST AND LOWEST SALARY PER ROLE

SELECT E.RID,MIN(S.SALARY),MAX(S.SALARY) FROM JOBROLES J
INNER JOIN EMPLOYEES E ON J.RID=E.RID INNER JOIN SALARIES S ON E.EMPID=S.EMPID GROUP BY E.RID;

Results | Messages

	RID	(No column name)	(No column name)
1	1	550000.00	980000.00
2	2	500000.00	935000.00
3	3	545000.00	985000.00
4	4	505000.00	970000.00
5	5	520000.00	995000.00

7.SALARY VS PERFORMANCE CORELATION

SELECT SUM((S.SALARY - AvgSalary) * (P.RATING - AvgRating)) /
SQRT(SUM(POWER(S.SALARY - AvgSalary, 2)) * SUM(POWER(P.RATING - AvgRating, 2))) AS CorrelationCoefficient FROM (SELECT S.SALARY, P.RATING, AVG(S.SALARY) OVER() AS AvgSalary, AVG(P.RATING) OVER() AS AvgRating FROM SALARIES S
JOIN PERFORMANCE_REVIEWS P ON S.EMPID = P.EMPID) AS Sub;

8.EMPLOYEES BELOW DEPARTMENT AVERAGE

SELECT E.EMPID, E.ENAME, D.DNAME AS Department, S.SALARY FROM EMPLOYEES E
JOIN DEPARTMENTS D ON E.DEPID = D.DEPID
JOIN SALARIES S ON E.EMPID = S.EMPID WHERE S.SALARY < (SELECT AVG(S2.SALARY) FROM EMPLOYEES E2
JOIN SALARIES S2 ON E2.EMPID = S2.EMPID WHERE E2.DEPID = E.DEPID);

Results | Messages

	EMPID	ENAME	Department	SALARY
1	103	Kiran Nair	HR	510000.00
2	107	Priya Sharma	HR	530000.00
3	112	Kavya Iyer	HR	555000.00
4	114	Amit Singh	HR	565000.00
5	123	Kavya Reddy	HR	610000.00
6	125	Swathi Sharma	HR	620000.00
7	126	Deepak Sharma	HR	625000.00
8	127	Rohan Reddy	HR	630000.00
9	138	Kavya Gupta	HR	685000.00
10	139	Meera Reddy	HR	690000.00
11	135	Neha Sharma	HR	670000.00

DEPARTMENT GROWTH

9.HIRING TREND PER YEAR

SELECT YEAR(JOINDATE) AS HireYear, COUNT(*) AS TotalHires FROM EMPLOYEES GROUP BY YEAR(JOINDATE) ORDER BY HireYear;

	HireYear	TotalHires
1	2015	9
2	2016	9
3	2017	15
4	2018	3
5	2019	9
6	2020	13
7	2021	8
8	2022	8
9	2023	11
10	2024	8
11	2025	7

10.HEAD COUNT GROWTH BY DEPARTMENTS

SELECT D.DNAME AS Department, YEAR(E.JOINDATE) AS YearJoined, COUNT(E.EMPID) AS Headcount FROM EMPLOYEES E
JOIN DEPARTMENTS D ON E.DEPID = D.DEPID GROUP BY D.DNAME, YEAR(E.JOINDATE) ORDER BY D.DNAME, YearJoined;

Results | Messages

	Department	YearJoined	Headcount
1	Data	2015	2
2	Data	2017	2
3	Data	2019	4
4	Data	2020	1
5	Data	2021	1
6	Data	2022	3
7	Data	2023	3
8	Data	2024	2
9	Data	2025	1
10	DevOps	2015	2
11	DevOps	2016	1

STRATEGIC INSIGHTS

12.EMPLOYEES OVER DUE FOR PROMOTION

SELECT E.EMPID, E.ENAME, D.DNAME AS Department, J.RNAME AS JobRole, E.JOINDATE, DATEDIFF(YEAR, E.JOINDATE, GETDATE()) AS YearsInCompany FROM EMPLOYEES E
JOIN DEPARTMENTS D ON E.DEPID = D.DEPID JOIN JOBROLES J ON E.RID = J.RID WHERE DATEDIFF(YEAR, E.JOINDATE, GETDATE()) >= 3;

Results | Messages

	EMPID	ENAME	Department	JobRole	JOINDATE	YearsInCompany
1	101	Arjun Sharma	DevOps	SDE 1	2021-03-29	5
2	102	Kiran Rao	Java	Data Scientist	2023-01-02	3
3	103	Kiran Nair	HR	SDE 1	2021-06-19	5
4	104	Rohan Kumar	Data	Data Scientist	2023-11-12	3
5	105	Manoj Rao	Data	Manager	2019-10-27	7
6	106	Neha Gupta	Java	Data Scientist	2015-10-30	11
7	107	Priya Sharma	HR	Data Scientist	2019-04-23	7
8	108	Rohan Gupta	DevOps	SDE 1	2022-11-28	4
9	109	Divya Rao	DevOps	Manager	2015-11-10	11
10	110	Divya Reddy	Data	Associate	2019-06-24	7
11	112	Kavya Iyer	HR	Manager	2016-05-13	10

13.HIGH SALARY BUT LOW PERFORMANCES CASES

SELECT E.EMPID, E.ENAME, D.DNAME AS Department, J.RNAME AS JobRole, S.SALARY, P.RATING FROM EMPLOYEES E
JOIN DEPARTMENTS D ON E.DEPID = D.DEPID
JOIN JOBROLES J ON E.RID = J.RID
JOIN SALARIES S ON E.EMPID = S.EMPID
JOIN PERFORMANCE_REVIEWS P ON E.EMPID = P.EMPID WHERE S.SALARY > (SELECT AVG(SALARY) FROM SALARIES) AND P.RATING < 3 ORDER BY S.SALARY DESC, P.RATING ASC;

	EMPID	ENAME	Department	JobRole	SALARY	RATING
1	199	Rohan Rao	Java	Manager	990000.00	2.8
2	198	Meera Rao	DevOps	Associate	985000.00	2.6
3	197	Naveen Gupta	DotNET	HR	980000.00	1.2
4	195	Swathi Rao	DevOps	Data Scientist	970000.00	1.6
5	194	Amit Kumar	DevOps	Data Scientist	965000.00	1.5
6	193	Priya Patel	Java	HR	960000.00	2.4
7	190	Kavya Iyer	Data	Data Scientist	945000.00	1.7
8	189	Amit Nair	DevOps	Associate	940000.00	1.4
9	188	Kiran Iyer	DevOps	SDE 1	935000.00	1.0
10	184	Pooja Iyer	Data	Manager	915000.00	2.5
11	179	Sneha Singh	DotNET	Associate	890000.00	1.8

16.LOCATION-WISE SALARY COMPARISION

SELECT L.LNAME AS Location, AVG(S.SALARY) AS AvgSalary, MIN(S.SALARY) AS MinSalary, MAX(S.SALARY) AS MaxSalary, COUNT(E.EMPID) AS EmployeeCount FROM EMPLOYEES E JOIN LOCATIONS1 L ON E.LID = L.LID JOIN SALARIES S ON E.EMPID = S.EMPID GROUP BY L.LNAME ORDER BY AvgSalary DESC;

	Location	AvgSalary	MinSalary	MaxSalary	EmployeeCount
1	Trivandrum	772000.000000	500000.00	995000.00	30
2	Hyderabad	743194.444444	505000.00	970000.00	36
3	Pune	730441.176470	515000.00	985000.00	34

17.GENDER DIVERSITY RATIO(IF ADDED)

SELECT D.DNAME AS Department, SUM(CASE WHEN E.GENDER = 'Male' THEN 1 ELSE 0 END) AS MaleCount, SUM(CASE WHEN E.GENDER = 'Female' THEN 1 ELSE 0 END) AS FemaleCount,

COUNT(E.EMPID) AS TotalEmployees, CAST(SUM(CASE WHEN E.GENDER = 'Female' THEN 1 ELSE 0 END) AS FLOAT) / NULLIF(COUNT(E.EMPID),0) AS FemaleRatio, CAST(SUM(CASE WHEN E.GENDER = 'Male' THEN 1 ELSE 0 END) AS FLOAT) / NULLIF(COUNT(E.EMPID),0) AS MaleRatio FROM EMPLOYEES E JOIN DEPARTMENTS D ON E.DEPID = D.DEPID GROUP BY D.DNAME ORDER BY FemaleRatio DESC;

Results | Messages

	Department	MaleCount	FemaleCount	TotalEmployees	FemaleRatio	MaleRatio
1	HR	7	11	18	0.611111111111111	0.388888888888889
2	DotNET	8	12	20	0.6	0.4
3	Data	8	11	19	0.578947368421053	0.421052631578947
4	Java	10	10	20	0.5	0.5
5	DevOps	15	8	23	0.347826086956522	0.652173913043478

18.MEDIAN SALARY CALCULATIONS

SELECT * FROM(SELECT SALARIES.*,ROW_NUMBER()OVER(ORDER BY SALARY ASC)AS R FROM SALARIES)AS SALNEW1 WHERE R=(SELECT ROUND(COUNT(*)/2,0)
FROM SALARIES);

Results | Messages

	SALID	EMPID	SALARY	EFFECTIVE_DATE	R
1	50	150	745000.00	2024-03-26	50

19.EMPLOYEES WITH MULTIPLE PROMOTIONS

SELECT E.EMPID, E.ENAME, COUNT(DISTINCT E.RID) AS rchanges FROM EMPLOYEES E JOIN JOBROLES J ON E.RID = J.RID GROUP BY E.EMPID, E.ENAME HAVING COUNT(DISTINCT E.RID) > 1 ORDER BY rchanges DESC;

20.IDENTIFY LEADERSHIP PIPELINE CANDIDATES

SELECT E.EMPID, E.ENAME, D.DNAME AS Department, J.RNAME AS JobRole, P.RATING, E.JOINDATE, DATEDIFF(YEAR, E.JOINDATE, GETDATE()) AS YearsInCompany, S.SALARY FROM EMPLOYEES E JOIN DEPARTMENTS D ON E.DEPID = D.DEPID
JOIN JOBROLES J ON E.RID = J.RID JOIN SALARIES S ON E.EMPID = S.EMPID
JOIN PERFORMANCE_REVIEWS P ON E.EMPID = P.EMPID WHERE P.RATING >= 4.5 AND DATEDIFF(YEAR, E.JOINDATE, GETDATE()) >= 3 AND E.STATUS = 'ACTIVE' ORDER BY P.RATING DESC, YearsInCompany DESC, S.SALARY DESC;

Results | Messages

	EMPID	ENAME	Department	JobRole	RATING	JOINDATE	YearsInCompany	SALARY
1	153	Amit Kumar	DevOps	SDE 1	5.0	2019-01-02	7	760000.00
2	108	Rohan Gupta	DevOps	SDE 1	5.0	2022-11-28	4	535000.00
3	171	Sneha Kumar	HR	HR	4.9	2017-10-06	9	850000.00
4	169	Rahul Gupta	DevOps	HR	4.8	2017-12-21	9	840000.00
5	192	Divya Rao	Data	Associate	4.8	2019-04-29	7	955000.00
6	134	Naveen Rao	DotNET	Data Scientist	4.8	2023-08-23	3	665000.00
7	117	Kavya Sharma	DevOps	Associate	4.6	2016-04-05	10	580000.00
8	180	Neha Rao	DotNET	SDE 1	4.5	2015-05-18	11	895000.00
9	173	Swathi Rao	DevOps	SDE 1	4.5	2017-12-25	9	860000.00
10	116	Swathi Rao	DotNET	Data Scientist	4.5	2017-05-11	9	575000.00